

AURA Hub — Technical Architecture Overview

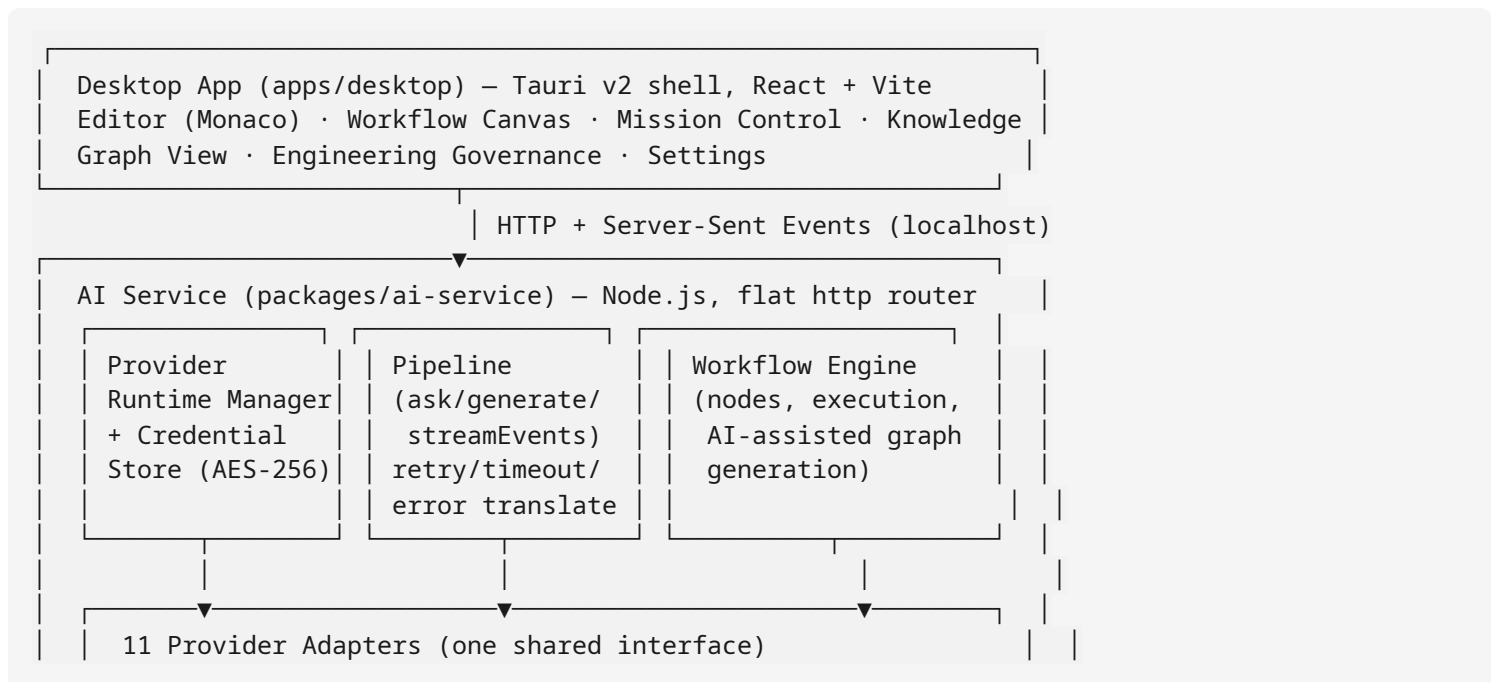
Prepared for technical partnership discussion with Alibaba Cloud.

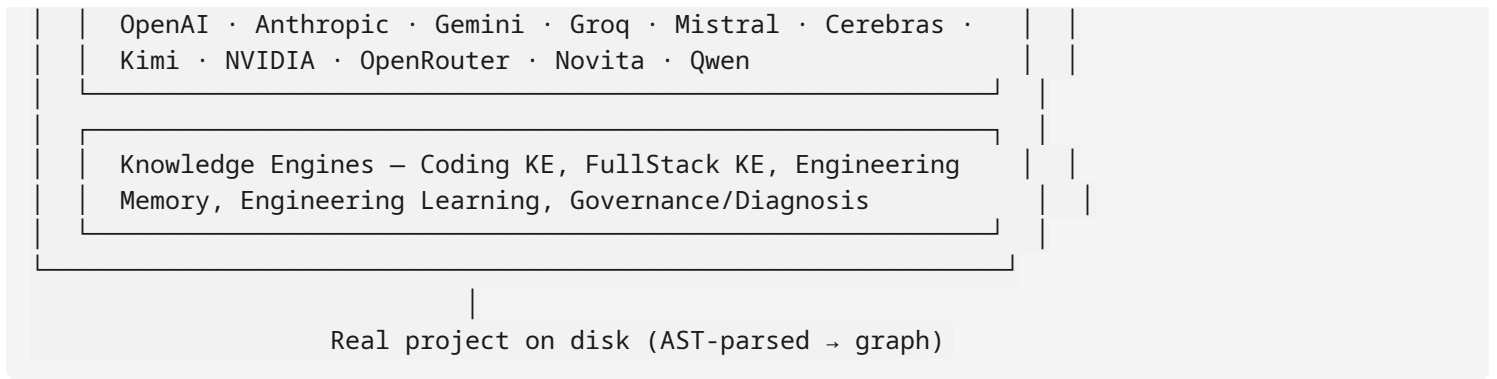
Repository: `Aura-hub-ai-native-workspace/aura-hub` (public, Apache-2.0).

1. What AURA Hub is

AURA Hub is an **AI-native engineering workspace** — a desktop application (Tauri v2 + React) paired with a local AI orchestration service (Node.js) that gives a developer a single environment combining: an AI assistant grounded in the real structure of their codebase, a visual workflow automation engine, multi-step autonomous task execution (Mission Control), a persistent engineering memory, and rule-based engineering governance (health scoring, diagnosis) — all running against the developer's own project, with **no built-in AI model**. Every AI capability is bring-your-own-key (BYOAK): the platform has no intelligence until the user connects a provider.

2. High-level architecture





3. The provider abstraction — why it matters for a partner

Every AI provider implements one interface:

```
interface ProviderAdapter {
  readonly metadata: ProviderMetadata;
  detect(apiKey: string): boolean;
  validate(apiKey: string): Promise<{ ok: boolean; error?: string }>;
  discoverModels(apiKey: string): Promise<DiscoveredModel[]>;
  createRuntime(apiKey: string, model?: string): Runtime;
  checkHealth(apiKey: string): Promise<ProviderHealth>;
}
```

Most OpenAI-compatible providers (including Qwen) need **zero method overrides** beyond declaring their base URL and default model — a shared base class (`BaseOpenAICompatible`) implements validation, model discovery, health checks, streaming, retry-eligible error classification, and timeout handling once. No feature, panel, or workflow node in the product ever branches on a provider id. This is a structural guarantee, not a convention: adding a new OpenAI-compatible provider is a single ~20-line file plus a one-line registry entry, and it is automatically available in every AI surface of the product — chat, inline code actions, the workflow builder, automation templates, mission planning, and engineering diagnosis — simultaneously.

This is the property that made integrating Qwen a clean, first-class addition rather than a bolt-on — see

[05-qwen-integration-overview.md](#) .

4. The workflow / automation engine

A workflow is a directed graph of typed nodes: source nodes (real project files, git diff, memory), intelligence nodes (the Coding/FullStack knowledge engines), generation nodes (routed through the active provider), logic nodes (condition/loop/delay), and action nodes (save memory, write a file, run a shell command, HTTP request). Nothing in a workflow is simulated — every node executes for real against the open project. The AI Workflow Builder turns a natural-language description into one of these graphs directly, using the same provider-agnostic generation path.

5. Knowledge and memory layers

- **Knowledge Graph** — an AST-derived structural graph of the codebase (god nodes, community detection, cross-file relationships), used to ground AI answers in real code rather than free-floating context.
- **Engineering Memory** — a persistent record of real engineering events (decisions, patterns, learnings), so the assistant's understanding compounds across sessions instead of resetting every conversation.
- **Engineering Governance** — deterministic, rule-based health scoring and drift detection layered on top of memory; the model's role is narrowed to explaining causes and drafting patches, not inventing scores.

6. Security & key handling

API keys are never sent to any AURA-operated server — there is none. Keys are encrypted at rest (AES-256-GCM) in a local store and only a fingerprint (first 4 / last 4 characters) is ever surfaced in the UI.

7. Current maturity — stated plainly

This is an early-stage, single-maintainer project (repository created 2026-07-28). The work described above is real and runs today, but it currently lives on a feature branch (`presentation-v0.1`) that has not yet been merged to `main` ; there is no hosted demo and no publicly captured screenshot set yet. We are raising this architecture for technical discussion on its merits, not on traction — see [01-partnership-proposal.md](#) for the honest framing of where this stands and what we're asking for.